

Storing Data in QR Codes

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Image Not Found

Figure 1: A human person engages in the typical QR (Quick Response) Code ritual of taking a picture of the mark and then visiting a web page (actions that most researchers consider to be linked through custom, rather than technology).

ABSTRACT

In the modern QR (Quadratic Residue) Code “scanning” ritual, the celebrant takes a photograph of a peculiar rectangular pattern using a smartphone before visiting a web page related to the item on which the rectangular pattern is inscribed. Some have conjectured that these two actions share a causal link, but this is clearly ridiculous. It is obviously much easier to find information on the internet using a simple text query instead of somehow entering a complete photograph, byte-by-byte, into a search engine.

What is research, however, if not structured engagement with the ridiculous? In this paper we develop techniques to store data in QR (Querulous Renegade) Codes without disrupting their aesthetic and ritualistic functionality.

CCS CONCEPTS

• **Technoethnography** → **End-User Semiotic Esotericism**; • **Applications** → **Web Programming**;

KEYWORDS

qr codes, data storage, information design

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1 INTRODUCTION

Modern society is full of fricative moments that serve no purpose, like sneezing at the sun, bidding farewell, and happiness. QR (Quite Rapid) Codes [Tikhonov 2019], and – particularly – the modern QR Code Ritual, are one such element.

QR (Quad Rectangle) Codes are square grids composed of contrasting light and dark squares. When encountering such a code, a believer will engage in the so-called QR (Quality Ritual) Code Ritual; performed as follows:

- (1) the practitioner notices a QR (Quiet Region) Code;
- (2) the practitioner photographs the code;
- (3) the practitioner visits a web page related to the item the code is affixed to.

While original ethnographies of the practice [unknown 2012] considered that steps 2 and 3 may be linked; subsequent scientific consensus has pushed back on this conclusion, arguing that, I mean, just *look* at a QR (Question Reaction) Code, dude (Figure 2). That’s random noise. Use your *common sense*: there is no data there.

Which, I mean, I guess so, *dude*.

2 METHOD

How, then, henceforth, and so on, can we adapt these pictographs to store data?

Particularly, we seek a method to add data to the random noise of the code without disrupting any ritual practice. Through many subtle interventions, we have determined that there are certain areas of each code which cannot be distorted without disrupting the ritual (Figure 3): the large squares in the corners (representing the centrality of the code within the cosmos), the smaller squares in the field (representing the multiplicity of experience of the viewer), the alternating-color lines between them (representing the bridges

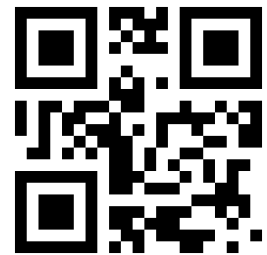


Figure 2: SETI@home all you want, it’s just random noise.

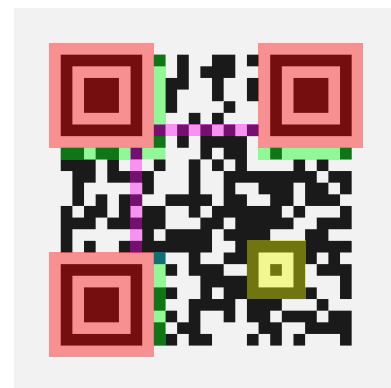


Figure 3: Regions of Extra Significance: I am he as you are he as you are me (red); and we are all together (purple); crabalocker (green); and the eggmen (yellow). The dark module is gently tinted in teal.

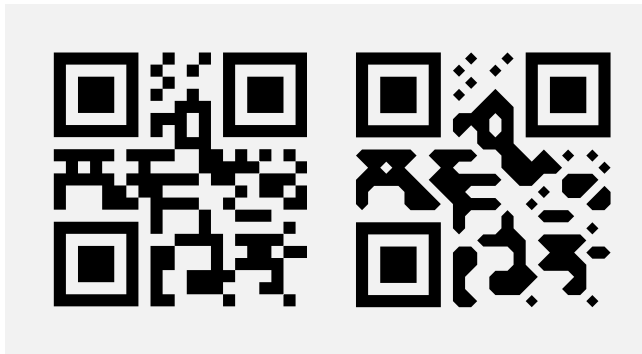


Figure 4: Cutting corners in a QR (Questionably Rectilinear) Code does not influence their ritual salience.

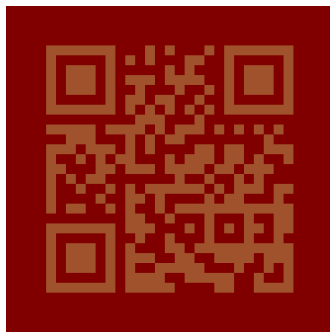


Figure 5: Using CSS extended color keywords allows conveying brief phrases in a QR (Quolor Region) Code.

of experience that can connect all things), particular cells adjacent to the large squares which vary between patterns (likely to appeal to the artistic sensibilities of one or more chaotic gods), and the so-called *dark module*, about which we must only speak in hushed tones.

Save for the squares, all of the other features have value to the ritual at their centers but appear to be adjustable around their edges. The ritual community appears to be tolerant of different code coloration as well, as long as that coloration is reasonably consistent within the pattern as a whole.

2.1 Cutting Corners

We always enjoy cutting corners in our research. Can we store data by cutting corners in a QR (Quorner Rectification) Code?

Of course! Our tests indicate that ritual practitioners are indifferent to whether a corner has been cut or not in their codes (Figure 4).

Thanks to extensive testing ($N = 1$) we conclude that QR (Quadruple Replication) Codes have 192 (± 0) corners; allowing one to store up to 190 bits of data (and two spares) by selectively cutting them.

2.2 Color

The QR (Quolor Region) Code ritual is largely invariant to choice of code color, as long as two contrasting tones are used. We can



Figure 6: Storing data in a QR (Quantized Relief) Code by using modules of different depth (left). The data can be recovered with a manual depth gauge (right), scrap paper, and an abundance of time.

leverage this free choice of contrasting foreground and background color to store data.

For digitally-represented codes, this gives us approximately 48 bits of encoding space. A bit less, because of our requirement of contrast.

For printed codes, it is unrealistic to expect exact color reproduction fidelity; but one can still make use of the CSS Colors Level 4 color names codebook to map colors to countries¹, fruits, flowers, rocks, and people² (Figure 5). Sort of a noun-heavy codebook, really.

Let's move on.

2.3 Depth

QR (Quickly Relenting) Code practitioners are obsessed with appearance over substance, and who can blame them?

We can.

Regardless, we can use this to slip some data into the QR Codes in 3D – we just make the darker parts holes of different depth (Figure 6). Then it's easy to read out a message with a depth gauge – no more than 10-15 minutes of work!

Depending on the depths to which you are willing to sink, this method provides an almost unlimited data storage capacity.

3 RESULTS

I've been showing you them throughout the paper. I'm not made of results, here. What, are you, some sort of evidence-wanting-person?

4 CONCLUSION

We've postulated three methods for storing information in QR (Quarrelsome Repetition) Codes – a heretofore entirely decorative art. In today's world, what with "AI" and all, postulating is basically the same as proving; just as thinking is the same as doing; watching is the same as doing; and wanting is the same as having.

REFERENCES

- Alexey Tikhonov. April 10, 2019. On Doluble-Sided QR-Codes. In *Proceedings of SIGBOVIK 2019*. ACH, Pittsburgh, PA, USA.
unknown. 2012. Pictures of People Scanning QR-codes. <https://picturesofpeoplescanningqrcodes.tumblr.com/>.

¹Well, a country.

²A few, specific, people.